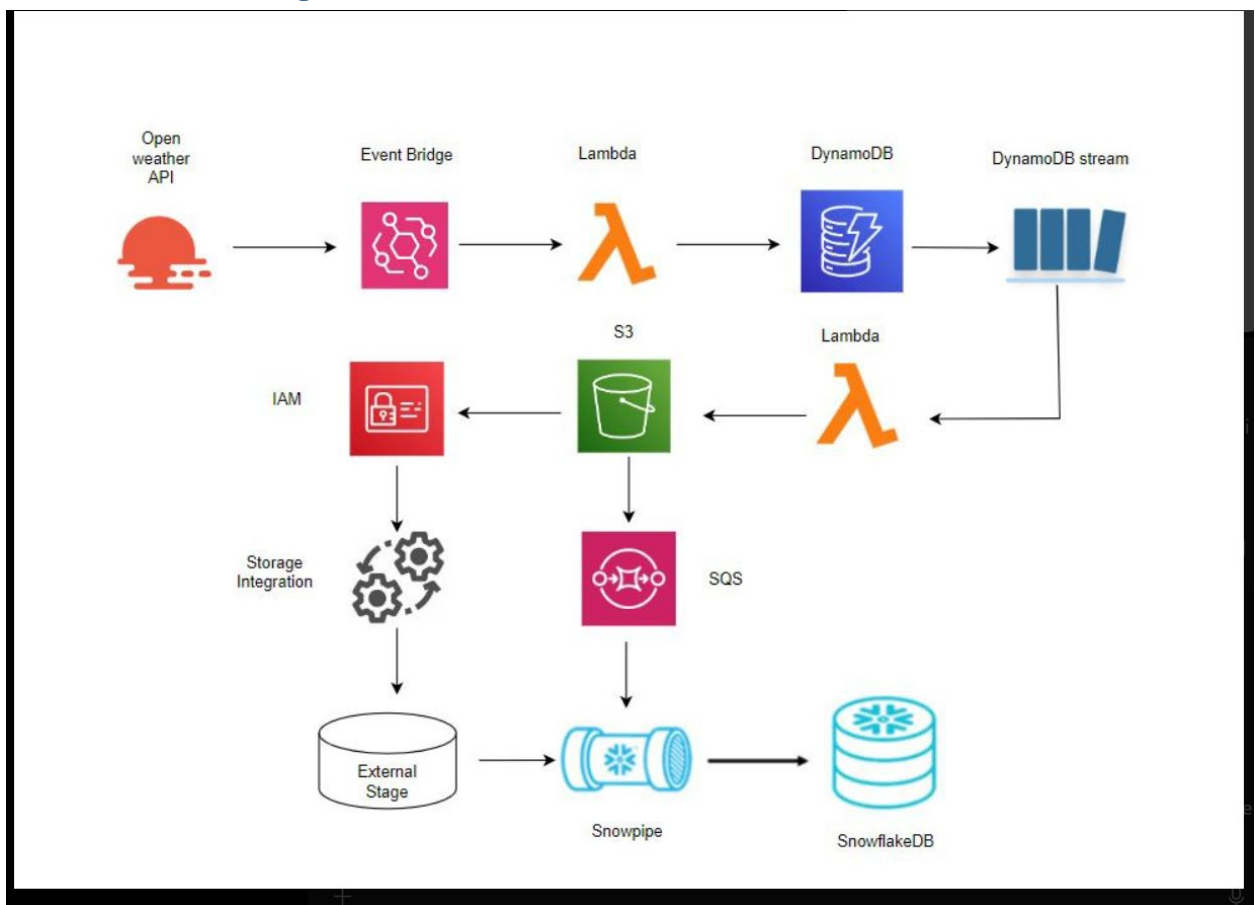


Weather Data Pipeline using AWS and Snowflake

1. Project Overview

This project implements an automated serverless weather data pipeline. It retrieves real-time weather information from the OpenWeather API, stores it in Amazon DynamoDB, cleans the data using Amazon S3, and automatically loads it into Snowflake using Snowpipe for analysis.

2. Architecture Diagram



3. Technologies Used

- OpenWeather API
- AWS Lambda
- Amazon EventBridge

- Amazon DynamoDB
- DynamoDB Streams
- Amazon S3
- Amazon SQS
- AWS IAM
- Snowflake
- Snowpipe
- Python

4. Pipeline Workflow

1. **Step 1: OpenWeather API**
A valid API key is used to fetch weather details such as city, temperature, humidity, weather condition, pressure and wind speed.
2. **Step 2: Amazon EventBridge**
Triggers the Lambda function automatically at scheduled intervals.
3. **Step 3: AWS Lambda**
Calls the OpenWeather API, extracts required fields and inserts them into a DynamoDB table.
4. **Step 4: Amazon DynamoDB**
Stores weather records in a structured table.
5. **Step 5: DynamoDB Streams**
Detects newly inserted records and triggers a second Lambda function.
6. **Step 6: Amazon S3**
The second Lambda writes data to S3, where unwanted fields, duplicate records and null values are removed. S3 acts as the staging and cleaned-data layer.
7. **Step 7: IAM & Storage Integration**
Provides secure access between Amazon S3 and Snowflake.
8. **Step 8: Amazon SQS**
Receives notifications when new files are uploaded to S3.
9. **Step 9: Snowpipe**
Automatically detects new files in S3 and loads them into Snowflake.
10. **Step 10: Snowflake**
Stores cleaned weather data for querying, reporting and visualization.

5. Sample Data

City: Kochi

Temperature: 31.5°C

Humidity: 78%

Weather: Clouds

Pressure: 1008 hPa

Wind Speed: 5.2 m/s

6. Conclusion

The pipeline automates weather data collection, data cleaning, and data loading into Snowflake. Using AWS serverless services minimizes operational overhead while ensuring scalable, near real-time analytics.